

Week 1 Internship Report

Networking Fundamentals & Network Simulation Tools

1. Introduction

This report covers the Week 1 task of the Network Administration Internship Program. The goal was to learn core networking fundamentals and get hands-on practice with two industry-standard simulation tools, Cisco Packet Tracer and GNS3, by building a simple LAN topology and testing connectivity with the ping command.

2. Networking Concepts Learned

- LAN (Local Area Network): a network of connected devices within a small area, such as an office.
- IP Address: a unique numerical identifier assigned to each device so it can be located and reached on a network.
- Subnet Mask: defines which portion of an IP address represents the network and which represents the individual device (e.g. 255.255.255.0).
- Switch: a device that connects multiple computers within a LAN and forwards data between them.
- Ping: a command used to test connectivity between two devices by sending small test packets and checking for a reply.

3. Cisco Packet Tracer — Interface Overview

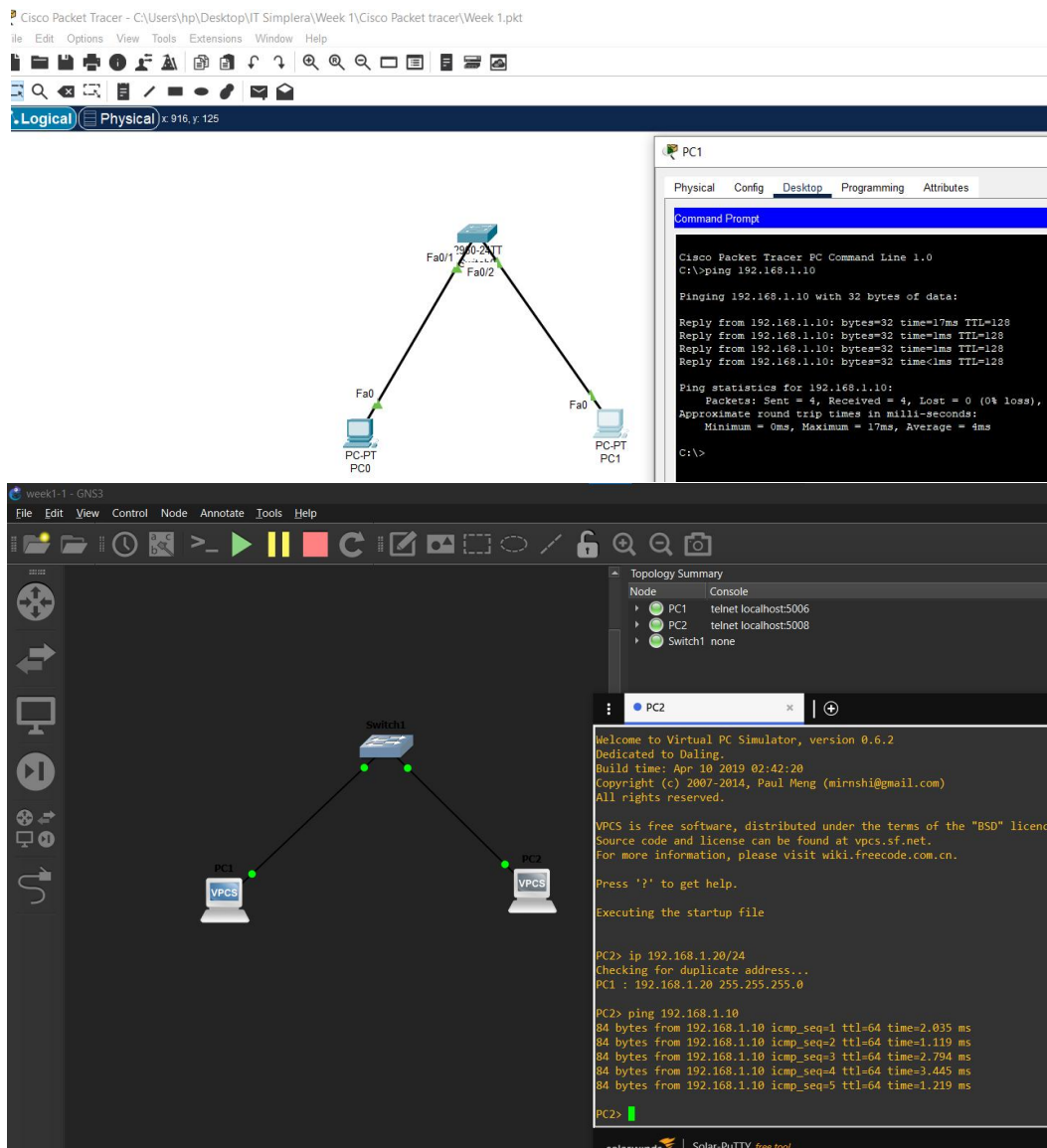
Packet Tracer provides a drag-and-drop workspace where network devices (PCs, switches, routers) can be placed and connected with virtual cables. Devices can be configured through simple GUI menus or a simulated command-line interface, and the tool supports both real-time and step-by-step simulation modes to observe how data moves through the network.

4. GNS3 — Interface Overview

GNS3 is a network simulator built around a project-based workspace. Devices are dragged onto a canvas from a device panel and linked using a cable tool. For this task, a Switch and two VPCS (Virtual PC Simulator) nodes were used, which are lightweight virtual PCs configured through a console/terminal rather than a graphical desktop.

5. Topology & Ping Test Results

A simple LAN was built in both tools consisting of two PCs connected through one switch. Static IP addresses were assigned (e.g. 192.168.1.10 and 192.168.1.20 on a 255.255.255.0 subnet), and the ping command was used to confirm successful communication between the two devices in both platforms.



6. Challenges Faced

The main challenge was using GNS3, since I was not familiar with the tool. Understanding the device panel, connecting nodes with the link tool, and figuring out how to start devices and configure IP addresses through the VPCS console took some time compared to Packet Tracer, which felt more beginner-friendly.

7. Conclusion

This task helped build a practical foundation in basic networking concepts and simulation tools. Successfully creating a LAN topology and verifying connectivity with ping in both Cisco Packet Tracer and GNS3 has improved my confidence in using these tools for future networking tasks.